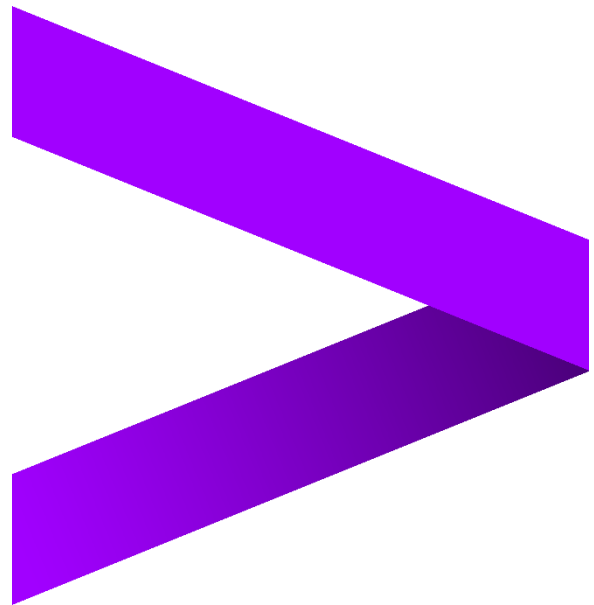


# **Data Analysis in Contour**

## **Lab Guide**



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# 1. Use case

## About the Company

TitaniumWorks Manufacturing is a company that produces high-quality industrial equipment. Like most manufacturing organizations, it relies heavily on machinery that must operate efficiently and safely. To maintain smooth operations, the company regularly inspects its equipment to prevent breakdowns and costly delays.

## The Challenge

The maintenance team is struggling to decide which machines should be inspected next. The process is currently manual and time-consuming, leading to inefficiencies and missed priorities. As a data analyst, you have been asked to step in and use Contour to bring data-driven insights into the decision-making process.

With Contour, you can:

- Save time by automatically re-running analyses when new data arrives.
- Build complex analyses easily through a visual, point-and-click interface.
- Create clear visualizations and reports that help non-technical teams understand results and act quickly.

## Your Goal

In this exercise, your main objective is to support TitaniumWorks in improving its equipment inspection strategy through data analysis. You will:

1. Clean and prepare the datasets for accurate analysis.
2. Combine multiple datasets to create a complete view of equipment performance.
3. Add calculated fields to reveal insights about maintenance schedules and identify high-risk machines.
4. Design visual dashboards to make the findings easy to interpret.
5. Embed your visualizations in a Notepad for seamless collaboration and presentation.

## Datasets

In this Lab, you'll be working with two datasets - one that stores equipment data and one that stores the production data of parts that are produced by the equipment.

- a. equipment.csv
- b. parts.csv

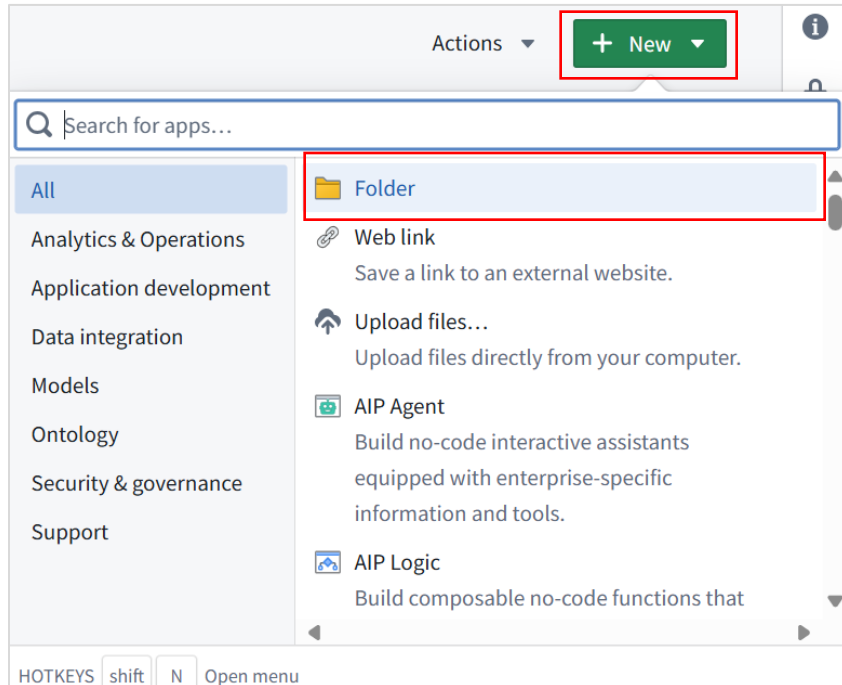
- If you already have your own Learning Project on your Foundry instance you can skip this step and move on to the next step to create your course-specific folder.

Navigate to Home Page -> Click on New Project -> Choose Basic Project under Template -> Provide a Name -> Click on Create Project

4 | Page

## b. Create a Folder

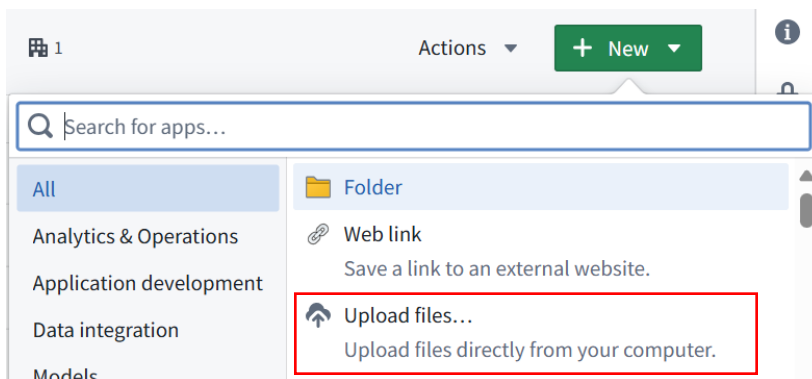
Open the Project folder -> Click on New -> Choose Folder -> Name the folder as Contour Deep Dive



| NAME ^                                                                                                | LAST UPDATED                | TAGS |
|-------------------------------------------------------------------------------------------------------|-----------------------------|------|
|  Contour Deep Dive | Tue, Oct 7, 2025, 7:51:1... |      |

## 3. Upload Data

a. Open the Contour Deep Dive Folder -> Click New -> Choose upload files



- b. Click on Choose from your computer -> Select the files -> Click on open -> Choose Upload as individual structured datasets -> click on Upload


Upload files



Drop files here or [choose from your computer](#)


equipment.csv
619 B


parts.csv
275.22 KB

☒
**Upload as individual structured datasets (recommended)**

Datasets are the most basic representation of tabular data. They can be used and transformed by many different applications. [Documentation](#)

☐
**Upload to a new media set**

Media sets enable media-specific capabilities for media files (e.g. audio, imagery, video, and documents). [Documentation](#)

☐
**Upload to a new unstructured dataset**

Unstructured datasets can store arbitrary files for processing and analysis. Structured data can be extracted from unstructured datasets using Pipeline Builder or Transforms. [Documentation](#)

☐
**Upload as individual raw files**

Raw files cannot be used in data pipelines, analyses, or models.

Upload


>

Contour Deep Dive

 1

Actions

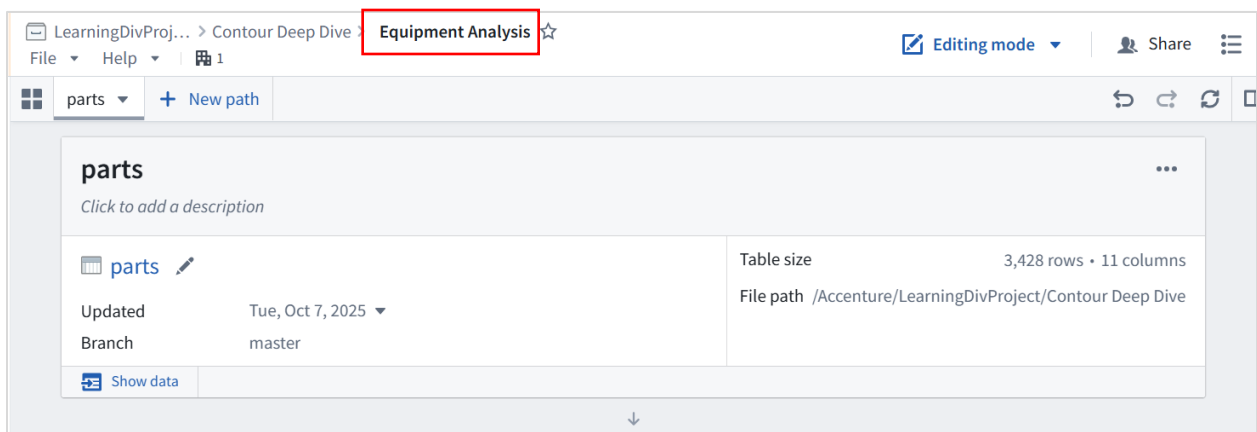
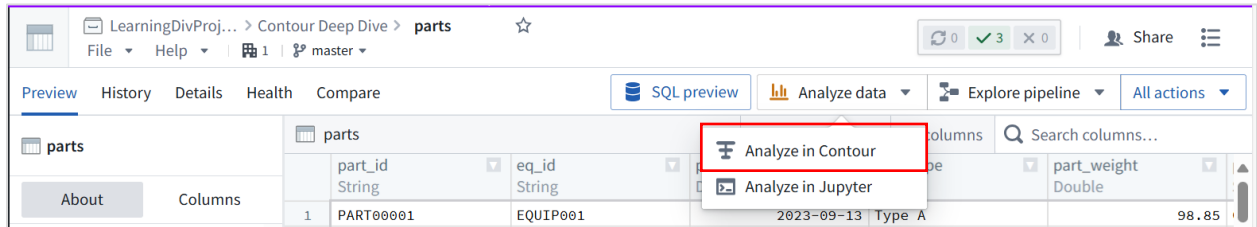
+ New

| NAME ^                                                                                        | LAST UPDATED                | TAGS |
|-----------------------------------------------------------------------------------------------|-----------------------------|------|
|  equipment | Tue, Oct 7, 2025, 6:42:3... |      |
|  parts     | Tue, Oct 7, 2025, 6:42:3... |      |

Upon successful uploading you can view the datasets under the Contour Deep Dive folder.

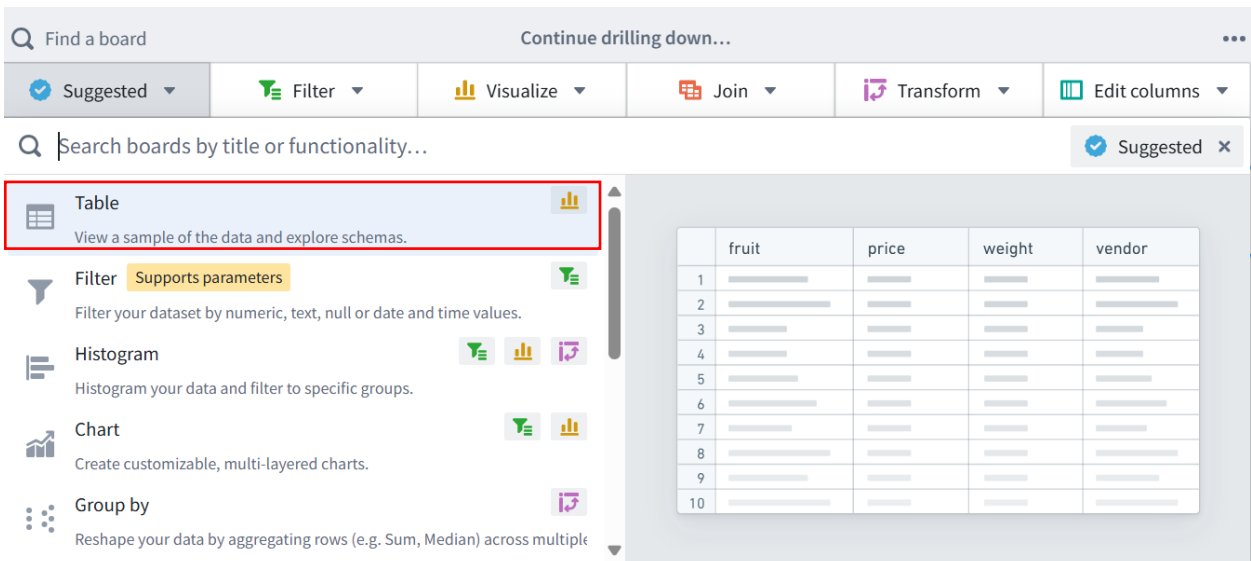
## 4. Create a Contour Analysis

- Select Parts file -> Click in Analyze Data -> Choose Analyze in Contour -> Rename the analysis as Equipment Analysis



## 5. Table Board

- In the board toolbar, click Suggested -> Table



Click on show data to display the data

parts + New path

Parts Data

Previewing 1,000 rows, 11 of 11 columns

|    | part_id<br>String | eq_id<br>String | part_producti...<br>Date | part_type<br>String | part_weight<br>Double | part_materi...<br>String |
|----|-------------------|-----------------|--------------------------|---------------------|-----------------------|--------------------------|
| 6  | PART00006         | EQUIP001        | 2024-05-24               | Type A              | 34.460                | Steel                    |
| 7  | PART00007         | EQUIP001        | 2024-05-18               | Type A              | 62.030                | Steel                    |
| 8  | PART00008         | EQUIP001        | 2024-03-19               | Type A              | 13.600                | Plastic                  |
| 9  | PART00009         | EQUIP001        | 2024-01-03               | Type A              | 23.710                | Plastic                  |
| 10 | PART00010         | EQUIP001        | 2023-08-25               | Type A              | 52.210                | Copper                   |
| 11 | PART00011         | EQUIP001        | 2024-01-30               | Type A              | 18.960                | Aluminum                 |
| 12 | PART00012         | EQUIP001        | 2023-11-23               | Type A              | 64.600                | Plastic                  |
| 13 | PART00013         | EQUIP001        | 2023-11-13               | Type A              | 82.090                | Plastic                  |
| 14 | PART00014         | EQUIP001        | 2023-09-19               | Type A              | 56.180                | Steel                    |
| 15 | PART00015         | EQUIP001        | 2024-05-07               | Type A              | 72.930                | Aluminum                 |
| 16 | PART00016         | EQUIP001        | 2023-09-23               | Type A              | 95.760                | Aluminum                 |
| 17 | PART00017         | EQUIP001        | 2023-10-20               | Type A              | 38.990                | Steel                    |
| 18 | PART00018         | EQUIP001        | 2023-09-22               | Type A              | 69.160                | Copper                   |
| 19 | PART00019         | EQUIP001        | 2023-11-05               | Type A              | 27.230                | Steel                    |
| 20 | PART00020         | EQUIP001        | 2023-11-07               | Type A              | 41.570                | Steel                    |
| 21 | PART00021         | EQUIP001        | 2023-11-14               | Type A              | 97.420                | Copper                   |
| 22 | PART00022         | EQUIP001        | 2024-05-03               | Type A              | 79.980                | Plastic                  |

Show data

## 6. Calculation Board

In the board toolbar, click Visualize

Calculation 1: Calculate the number of unique parts by choosing the

Column: part\_id

Aggregate: Unique count

Calculation 2: Calculate the number of unique equipment ID by choosing the

Column: eq\_id

Aggregate: Unique count

Unique counts based on Part\_id and eq\_id

Add to dashboard

Data Format

+ Add calculation

Unique count of part\_id ^ v x

Unique count of eq\_id ^ v x

Compute

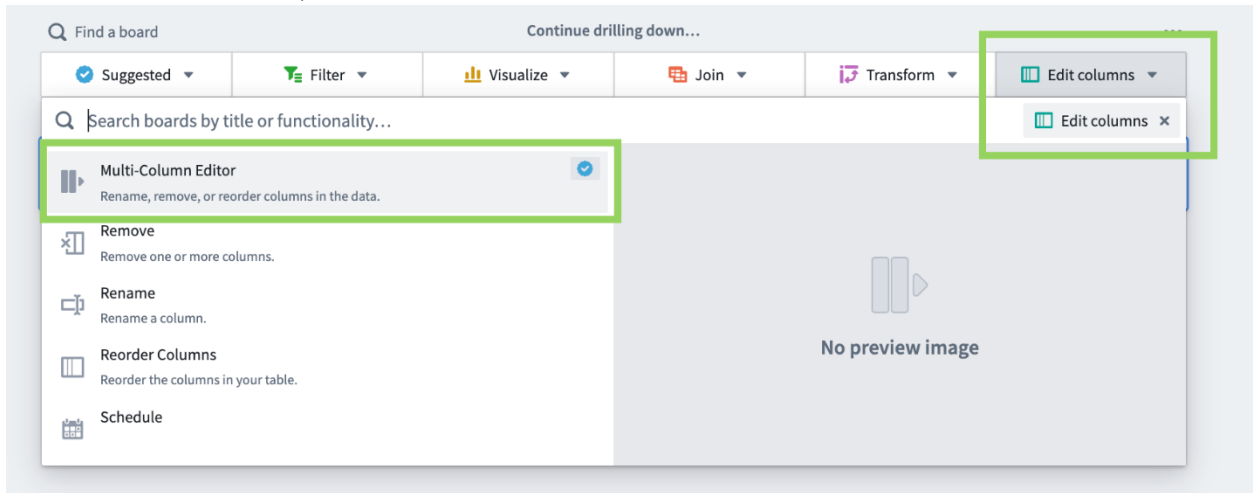
Show data

|                         |                       |
|-------------------------|-----------------------|
| Unique count of part_id | Unique count of eq_id |
| 3,428                   | 5                     |

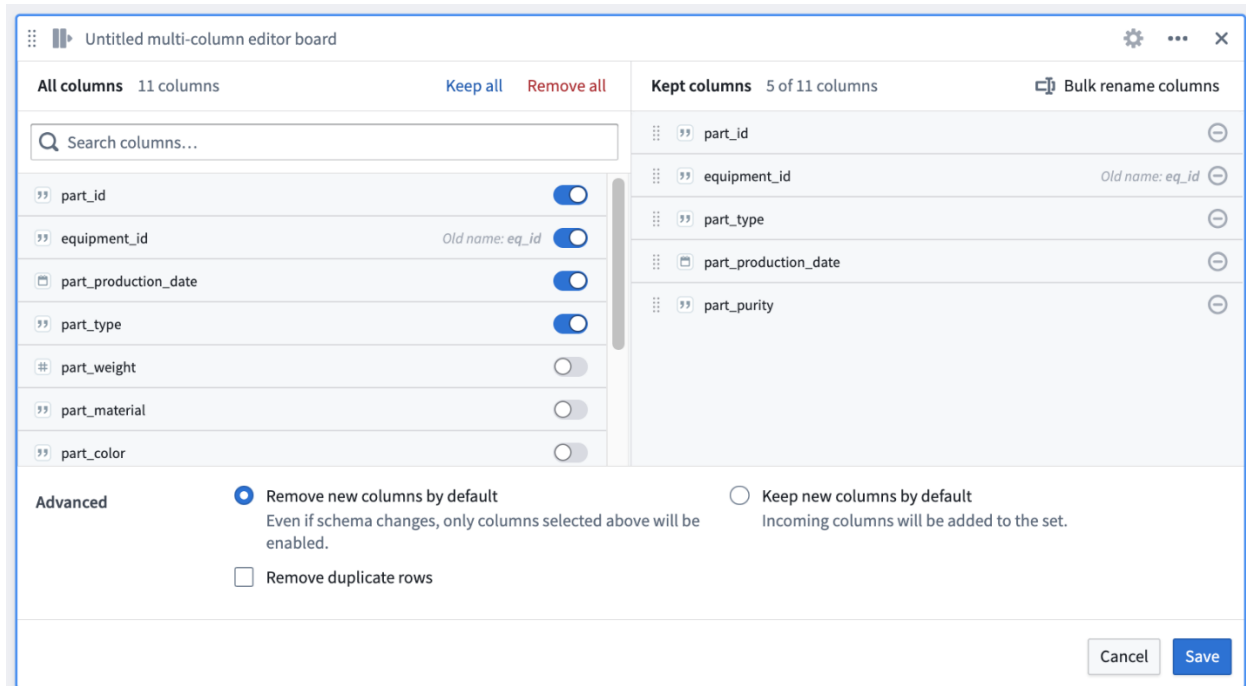


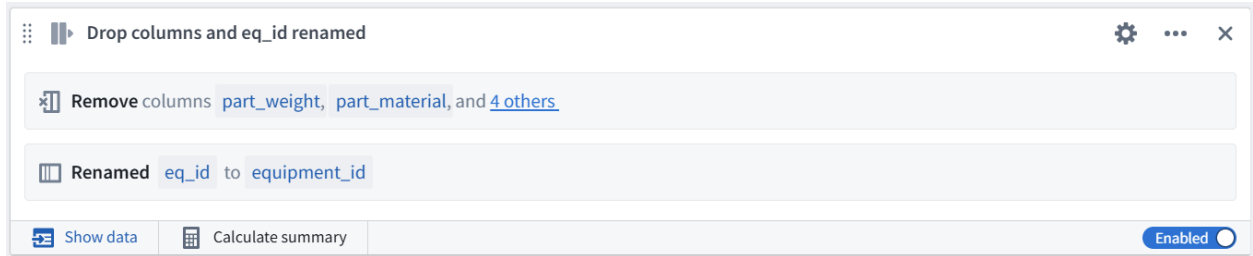
## 7. Rename and Drop Columns

- In the board toolbar, Edit columns > Multi-Column Editor



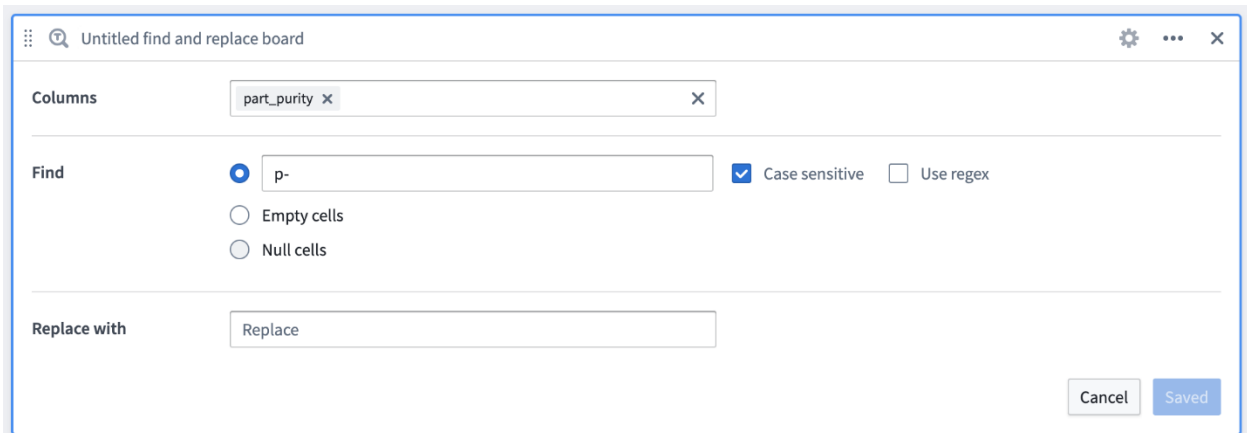
- In the right Kept columns panel, click on the eq\_id column -> rename the column to equipment\_id
- Under all columns, toggle off the below columns and click apply / save
  - part\_weight
  - part\_material
  - part\_color
  - part\_dimensions
  - part\_cost
  - part\_quality\_grade





## 8. Find and Replace Column Board

- In the board toolbar, **Transforms > Find and Replace** -> Remove the p- prefix from part\_purity using below options and click on Apply
  - Columns: part\_purity
  - Find: p-
  - Replace: (leave this empty)



| part_id<br>String | equipment_id<br>String | part_productio...<br>Date | part_type<br>String | part_purity<br>String |
|-------------------|------------------------|---------------------------|---------------------|-----------------------|
| PART00001         | EQUIP001               | 2023-09-13                | Type A              | 0.95                  |
| PART00002         | EQUIP001               | 2023-07-10                | Type A              | 0.941                 |
| PART00003         | EQUIP001               | 2024-04-22                | Type A              | 0.945                 |
| PART00004         | EQUIP001               | 2024-03-21                | Type A              | 0.946                 |
| PART00005         | EQUIP001               | 2024-01-21                | Type A              | 0.952                 |

## 9. Convert Types Board

- In the board toolbar, **Transform > Convert types** -> convert the part\_purity to Double by using the below options and click on Apply
  - Type conversions: Select column > part\_purity
  - On the right, change String to Double

parts purity

Convert type of `part_purity` to Double

Show data
Calculate summary
Enabled

| part_id<br>String | equipment_id<br>String | part_productio...<br>Date | part_type<br>String | part_purity<br>Double |
|-------------------|------------------------|---------------------------|---------------------|-----------------------|
| PART00001         | EQUIP001               | 2023-09-13                | Type A              | 0.95                  |
| PART00002         | EQUIP001               | 2023-07-10                | Type A              | 0.941                 |
| PART00003         | EQUIP001               | 2024-04-22                | Type A              | 0.945                 |
| PART00004         | EQUIP001               | 2024-03-21                | Type A              | 0.946                 |
| PART00005         | EQUIP001               | 2024-01-21                | Type A              | 0.952                 |

## 10. Filter Board

- In the board toolbar, **Filter > Filter**. Set Filter 1 to keep only production dates in 2024 using the below options and click Apply.
  - Select column: `part_production_date`
  - Comparison: on or after
  - Date: Select Jan 1, 2024

Untitled filter board

Keep rows that match all filters

Filter 1
part\_production\_date
on or after
Jan 1, 2024

Add a filter

Remove duplicate rows

Find a board
Continue drilling down
Suggested
Filter
Visualize

Edit columns

Untitled filter board

Keep rows where part\_production\_date is on or after Jan 1, 2024

Show data
2,176 rows • 5 columns
Enabled

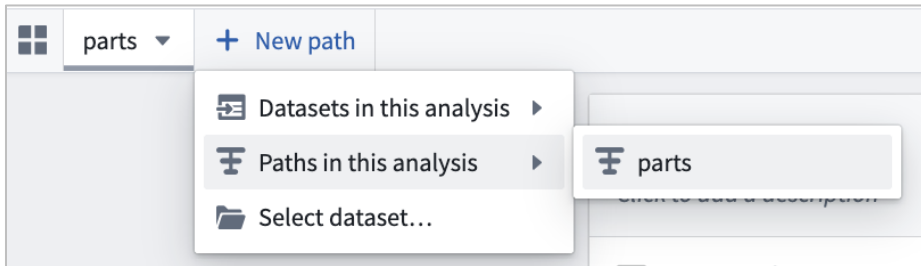
Click **Calculate summary** in the lower left of your new filter board

- After calculation, you should see a row count of 2,176 rows. All subsequent boards will use this filtered data.

## 11. Joining Datasets

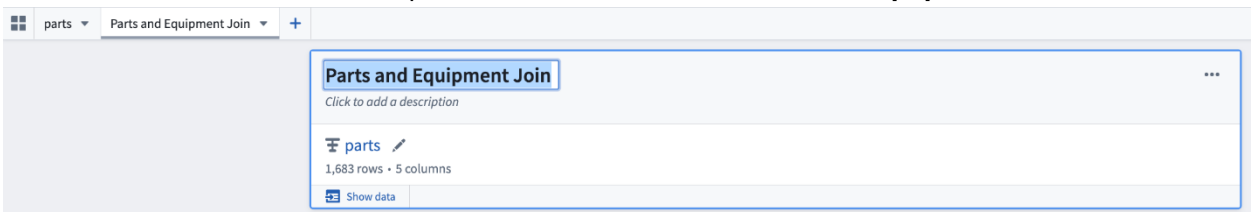
### a. Add New Path to the Analysis

- Click on the new path at the top of the analysis -> Choose Paths in the analysis -> parts



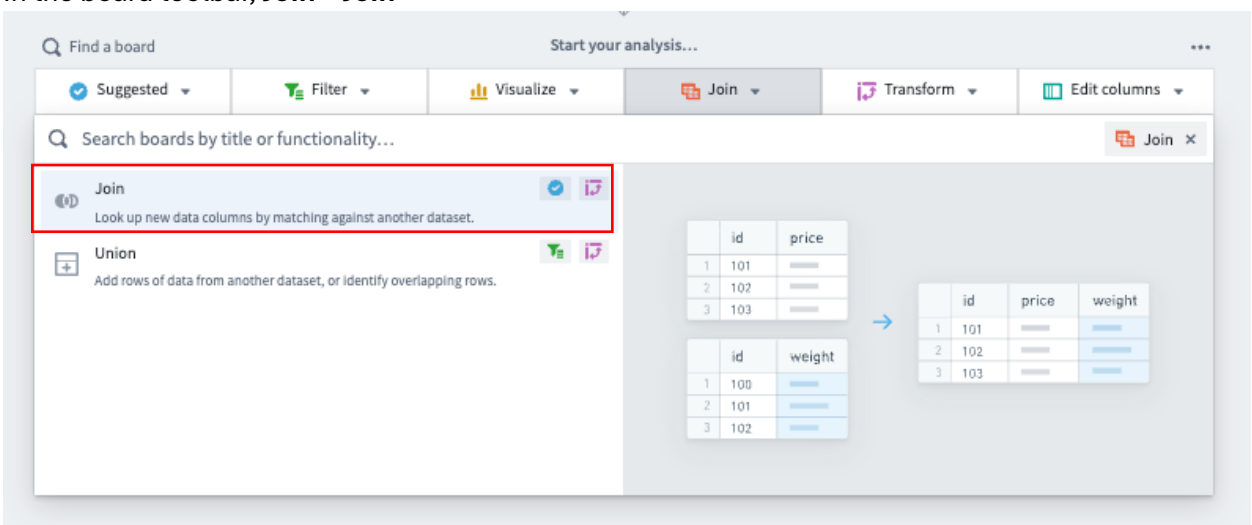
In the new tab, the first input board will display **parts (resulting set)** as the title. This means the output of the **parts** path will flow as input into this new path

- Double click on the title of the input board -> Rename it to **Parts and Equipment Join**

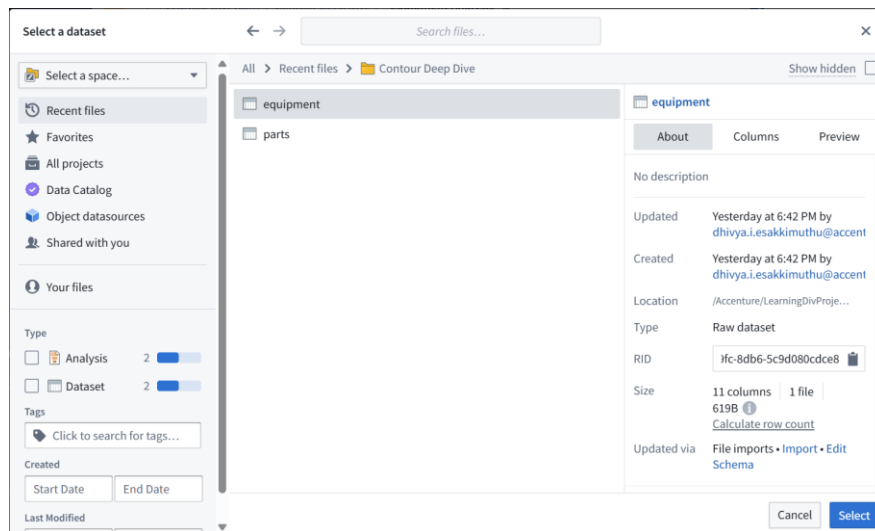


### b. Joining Datasets

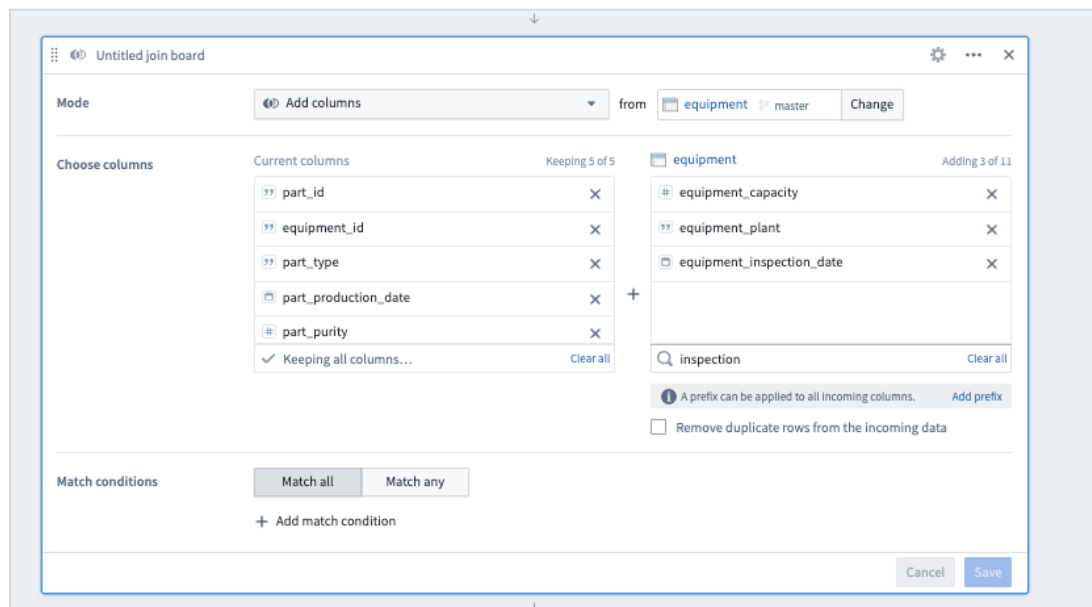
- In the board toolbar, **Join > Join**



- Add columns from the equipment dataset
  - Leave the Mode set to Add columns
  - Click Choose dataset... -> Select dataset
  - Search for and select the equipment dataset that you previously uploaded to your course folder



- In the equipment field on the right side of the window, select Add columns to add
  - equipment\_capacity, equipment\_plant, equipment\_inspection\_date
- *Warning: There is both an installation and an inspection date.*



- Add Matching condition as below and click on Save

Match conditions

Match all Match any

equipment\_id ↔ equipment\_id

+ Add match condition

Cancel Save

Click on show data to display the data.

| equipment_id<br>String | part_productio...<br>Date | part_type<br>String | part_purity<br>Double | equipment_ca...<br>Integer | equipment_plant<br>String |
|------------------------|---------------------------|---------------------|-----------------------|----------------------------|---------------------------|
| EQUIP001               | 2024-04-22                | Type A              | 0.945                 | 1200                       | Plant 1                   |
| EQUIP001               | 2024-03-21                | Type A              | 0.946                 | 1200                       | Plant 1                   |
| EQUIP001               | 2024-01-21                | Type A              | 0.952                 | 1200                       | Plant 1                   |
| EQUIP001               | 2024-05-24                | Type A              | 0.954                 | 1200                       | Plant 1                   |
| EQUIP001               | 2024-05-18                | Type A              | 0.943                 | 1200                       | Plant 1                   |

## 12. Expression Board

- In the board toolbar, Transform > Expression -> Click the Editor tab

Suggested Filter Visualize Join Transform Edit columns

Search boards by title or functionality...

Convert types  
Cast one or more columns to a new data type.

Export  
Export path results to a CSV or XLS.

**Expression** Supports parameters  
Use the expression language to derive new columns or perform complex fi

Find and Replace  
Replace values within one or more columns.

Group by

Library Editor

Expression 1

Order status code

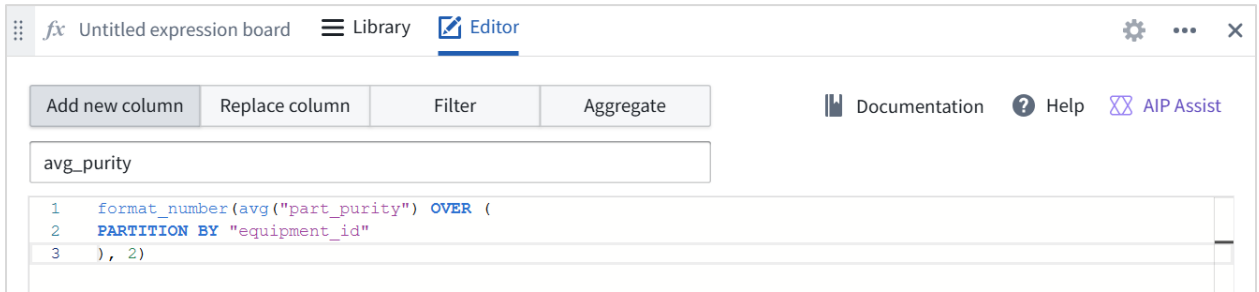
Expression 3

```

1 case
2 when ( " " ) then -1
3 when ( " ")>= 0 and
4 ( " " ) then 1
5 else
6 end

```

- Create a new column using the expression below -> click on save  
`format_number(avg("part_purity") OVER (PARTITION BY "equipment_id", 2)`



Click on show data to verify the expression value.

| avg_purity<br>String | part_id<br>String | equipment_id<br>String | part_productio...<br>Date | part_type<br>String | part_purity<br>Double |
|----------------------|-------------------|------------------------|---------------------------|---------------------|-----------------------|
| 0.95                 | PART00003         | EQUIP001               | 2024-04-22                | Type A              | 0.945                 |
| 0.95                 | PART00004         | EQUIP001               | 2024-03-21                | Type A              | 0.946                 |
| 0.95                 | PART00005         | EQUIP001               | 2024-01-21                | Type A              | 0.952                 |
| 0.95                 | PART00006         | EQUIP001               | 2024-05-24                | Type A              | 0.954                 |
| 0.95                 | PART00007         | EQUIP001               | 2024-05-18                | Type A              | 0.943                 |

## 13. Expression board for Case

- Use another Expression board to generate inspection alerts -> Add another **Expression** board (**Transform > Expression**) -> Under Editor tab -> Add New column named **inspection\_alert**.
- Use the below case expression and click on save

**CASE**

WHEN date\_diff(current\_date(), "equipment\_inspection\_date") > 365

AND "avg\_purity" < .90 THEN 'high'

WHEN date\_diff(current\_date(), "equipment\_inspection\_date") > 365

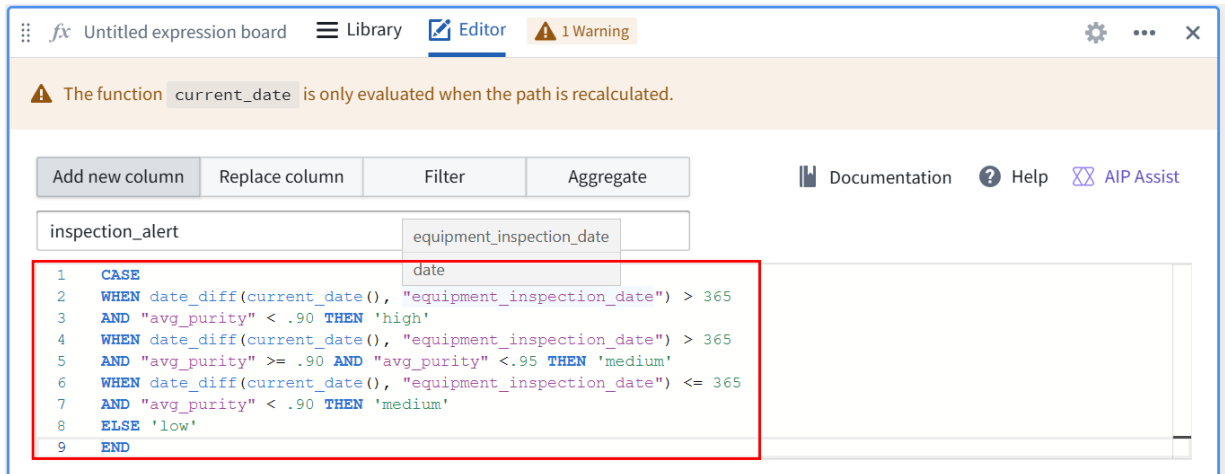
AND "avg\_purity" >= .90 AND "avg\_purity" < .95 THEN 'medium'

WHEN date\_diff(current\_date(), "equipment\_inspection\_date") <= 365

AND "avg\_purity" < .90 THEN 'medium'

ELSE 'low'

**END**



The function `current_date` is only evaluated when the path is recalculated.

Buttons: Add new column, Replace column, Filter, Aggregate

Documentation, Help, AIP Assist

```

1 CASE
2   WHEN date_diff(current_date(), "equipment_inspection_date") > 365
3   AND "avg_purity" < .90 THEN 'high'
4   WHEN date_diff(current_date(), "equipment_inspection_date") > 365
5   AND "avg_purity" >= .90 AND "avg_purity" < .95 THEN 'medium'
6   WHEN date_diff(current_date(), "equipment_inspection_date") <= 365
7   AND "avg_purity" < .90 THEN 'medium'
8   ELSE 'low'
9 END

```

Click on show data to preview the results

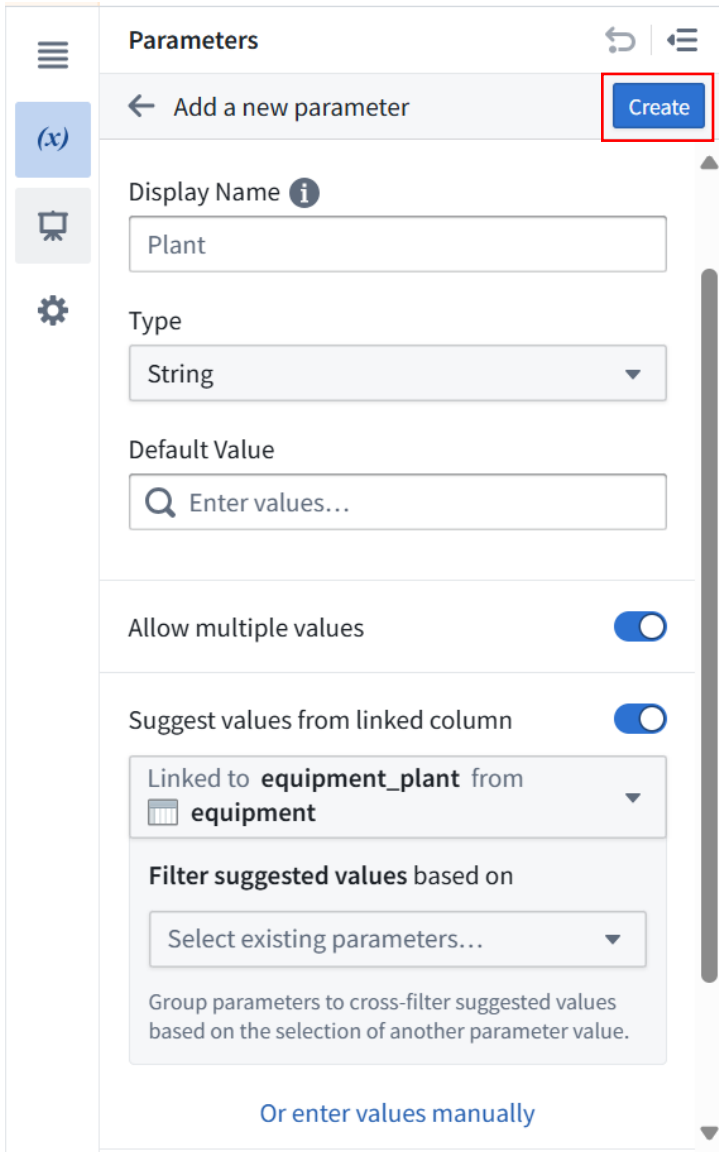
| inspection_alert<br>String | avg_purity<br>String | part_id<br>String | equipment_id<br>String | part_productio...<br>Date | part_type<br>String |
|----------------------------|----------------------|-------------------|------------------------|---------------------------|---------------------|
| low                        | 0.95                 | PART00681         | EQUIP001               | 2024-04-19                | Type A              |
| low                        | 0.95                 | PART00682         | EQUIP001               | 2024-03-30                | Type A              |
| low                        | 0.95                 | PART00683         | EQUIP001               | 2024-02-03                | Type A              |
| high                       | 0.89                 | PART00685         | EQUIP002               | 2024-01-06                | Type B              |
| high                       | 0.89                 | PART00688         | EQUIP002               | 2024-03-05                | Type B              |

## 14. Parameters

### a. Create a parameter

- Click on the **Parameters** icon (x) from the left-hand sidebar
- Click the + Create parameter button at the center of the left panel.
- Configure the parameter as below and click on create





**Parameters**

← Add a new parameter **Create**

Display Name ⓘ  
Plant

Type  
String

Default Value  
Enter values...

Allow multiple values ☒

Suggest values from linked column ☒

Linked to **equipment\_plant** from **equipment**

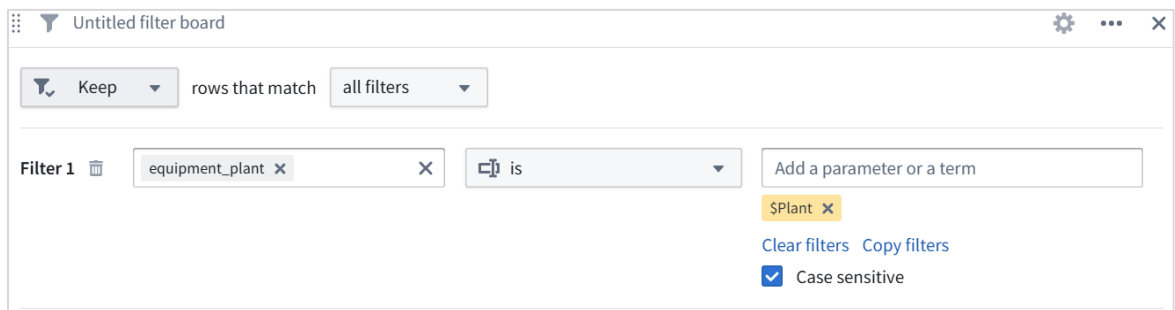
Filter suggested values based on  
Select existing parameters...

Group parameters to cross-filter suggested values based on the selection of another parameter value.

[Or enter values manually](#)

## b. Parameter in filter

- From the board toolbar, **Filter > Filter**
- Define as filter below with the parameter and click on save



Untitled filter board

Keep rows that match all filters

Filter 1 **equipment\_plant** is **SPlant**

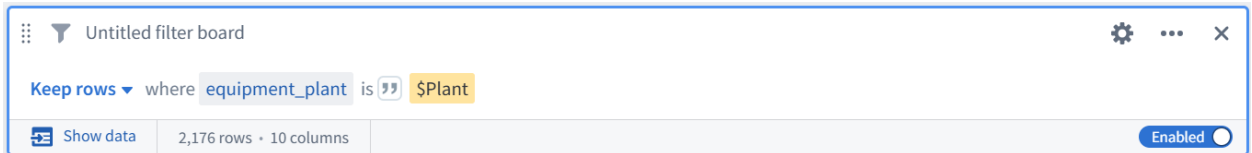
Add a parameter or a term

[Clear filters](#) [Copy filters](#)

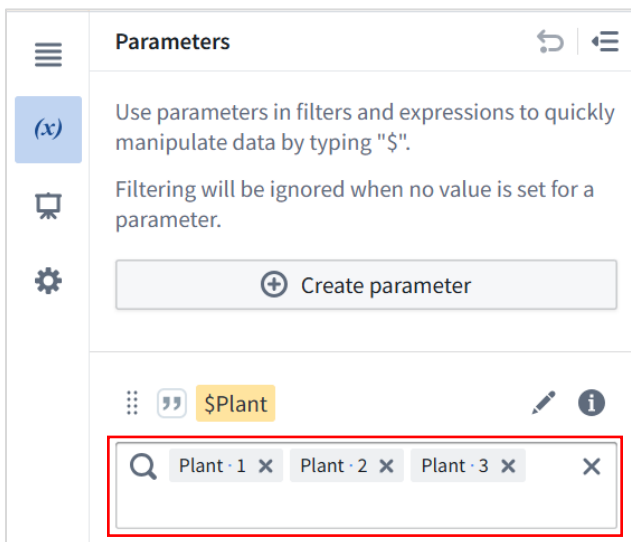
☒ Case sensitive

### c. Pass values to Parameters

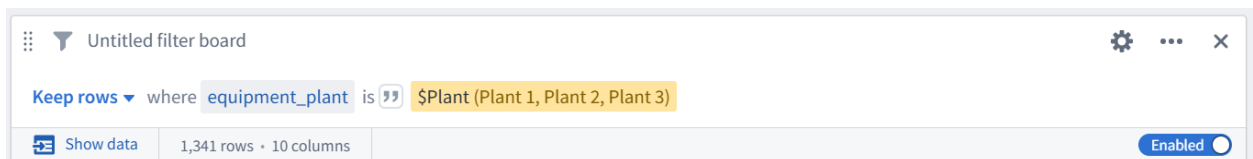
- Click Calculate summary in the lower left corner of the two most recent expression boards
  - Notice that the datasets are the same at this point. You have not defined the value for the \$plant parameter, so no filtering is being done.



- Open the left-side **Parameters** panel -> Enter values for **\$plant**: Plant 1, Plant 2, and Plant 3
- Click **Apply** at the bottom right of the Parameters panel



- Now click **Calculate summary** on your latest filter board
- The row count should update to 1,341 as you are now filtering only to rows containing data for Plant 1, Plant 2, and Plant 3.



Clear the parameters value in the parameter panel and click on apply before proceeding with the next step

## 15. Visualizations

### a. Histogram

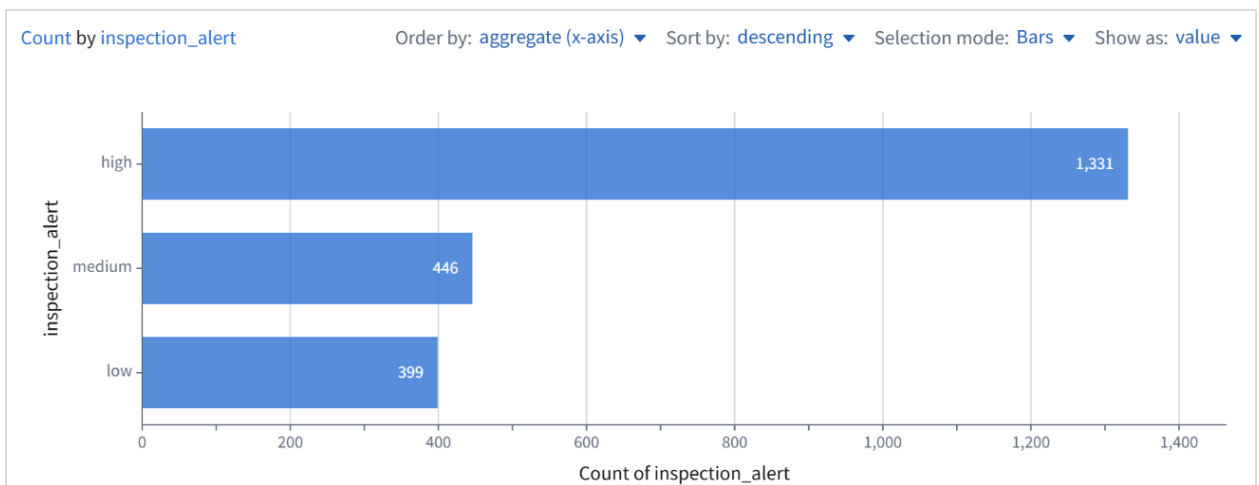
- In the board toolbar, **Visualize > Histogram** -> Configure the board to show a count of inspection alerts using the below options and click on compute
  - Under Y-Axis, Column: inspection\_alert
  - Under X-Axis, select Count

Configure Board

Y-Axis  
 Column inspection\_alert

X-Axis  
 Count of Select column...

Close Compute

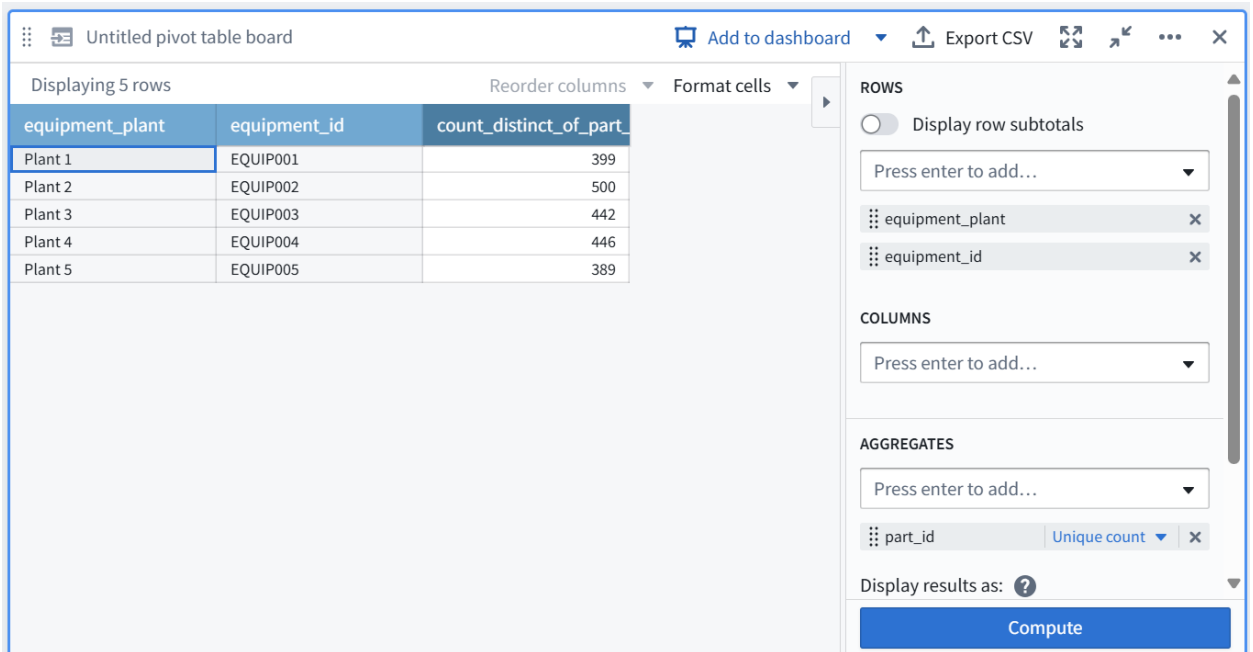


Click on the bar representing high values.

A grey bar at the bottom of the board will appear displaying, "Keep rows where inspection\_alert is high." This will filter the data in all subsequent boards.

## b. Pivot Table

- In the board toolbar, **Transform > Pivot Table** -> Configure the pivot table as below and click on compute
  - Rows: Add equipment\_plant and equipment\_id
  - Columns: Leave empty
    - While you have the option to select a column for a deeper analysis of each aggregated row, we'll leave it blank for this simple analysis.
  - Aggregates: Select part\_id and ensure that Unique count is selected.

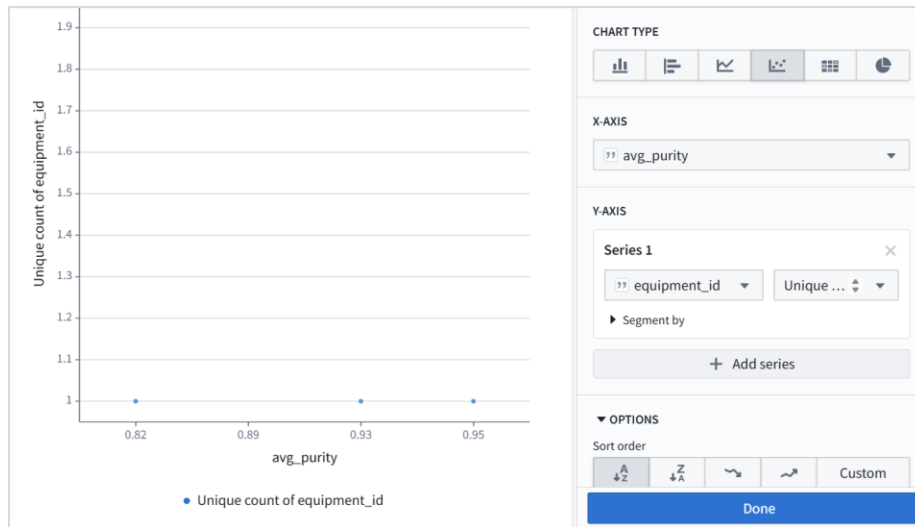


| equipment_plant | equipment_id | count_distinct_of_part |
|-----------------|--------------|------------------------|
| Plant 1         | EQUIP001     | 399                    |
| Plant 2         | EQUIP002     | 500                    |
| Plant 3         | EQUIP003     | 442                    |
| Plant 4         | EQUIP004     | 446                    |
| Plant 5         | EQUIP005     | 389                    |

## c. Chart – Scatter plot

- In the board toolbar, **Visualize > Chart** -> Configure the chart as below and click on save
  - Chart Type: Scatter plot
    - If you hover your cursor over each option, the tooltip will tell you the title of each one.
    - Note that the same Chart board can produce many different kinds of charts, such as bar, line, scatter, heat grid and pie charts.
  - X-Axis: avg\_purity.
  - Y-Axis: Series 1: Select equipment\_id and Unique count
  - Options: Set Sort order to X axis ascending

Again, you can hover over each option to reveal its title.



#### d. Chart – Vertical Bar Chart

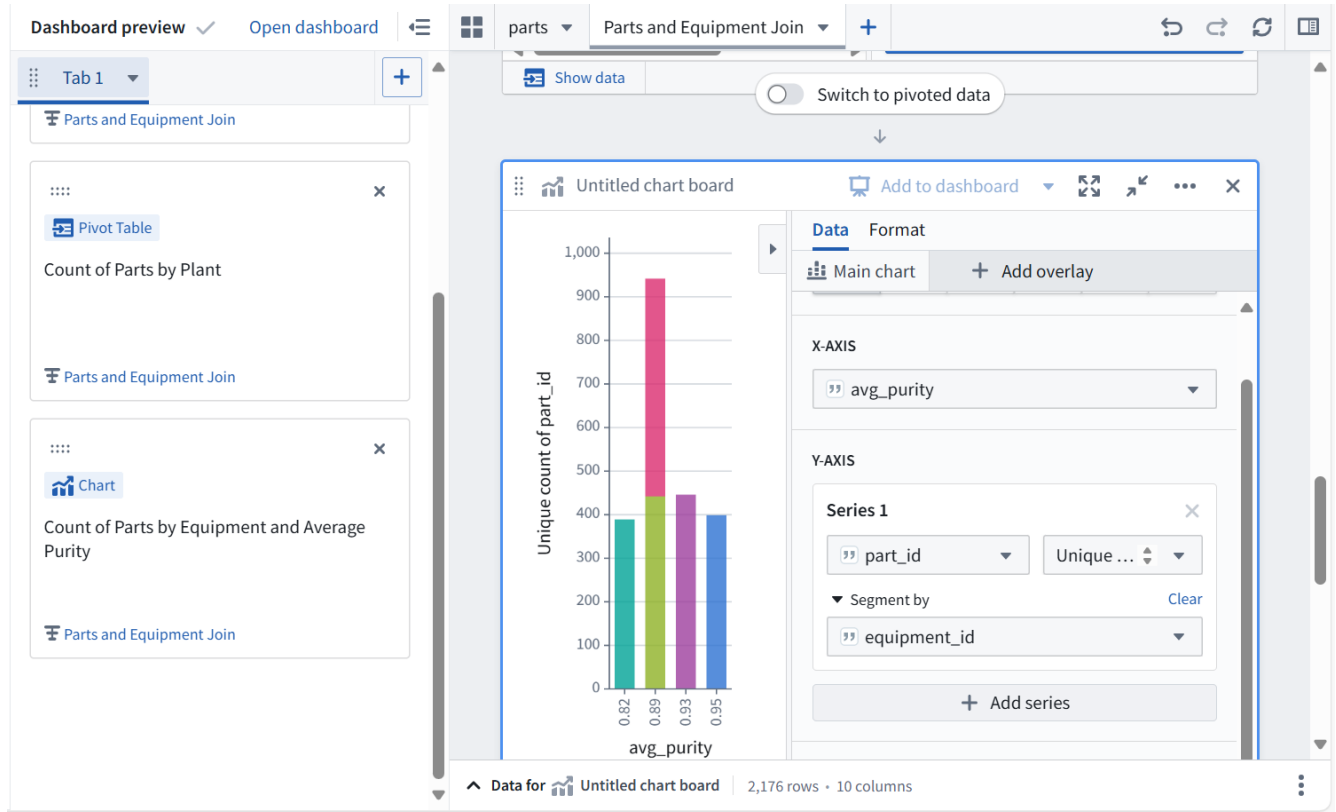
- In the configuration of your existing chart / add a new chart board in the path and configure as follows and click on compute / save
  - Chart type: **Vertical bar chart**
  - X-Axis: avg\_purity
  - Y-Axis: Series 1: Select part\_id and **Unique count**
  - Segment by: equipment\_id
  - Options: Segment options: **Stacked**



## 16. Contour Dashboard

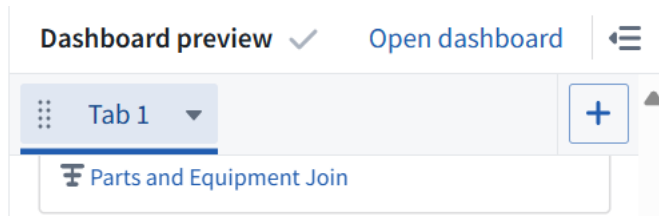
### a. Create a dashboard of your analysis

1. Scroll to your Histogram board
2. Click **Add to dashboard** at the top of that board
  - The dashboard preview panel will appear on the left side of your screen and a blue *Dashboard* button will appear at the top.
3. Title this board: Inspection Alerts Histogram
  - The box in the center of the Histogram card in the dashboard preview allows you to title this board within the dashboard.
4. Add a title to your dashboard
  - Click into the text box at the top of the dashboard preview, which says *Untitled dashboard*
  - Enter: Inspection Prioritization Dashboard
5. Add your pivot table to the dashboard
  - Scroll to your pivot table board
  - Click **Add to dashboard**
  - Title the board: Count of Parts by Plant
6. Add your bar chart to the dashboard
  - Scroll to your pivot table board
  - Click **Add to dashboard**
  - Title the board: Count of Parts by Equipment and Average Purity



## b. Interact with Dashboard

- Click **Open dashboard** at the top of the left dashboard panel



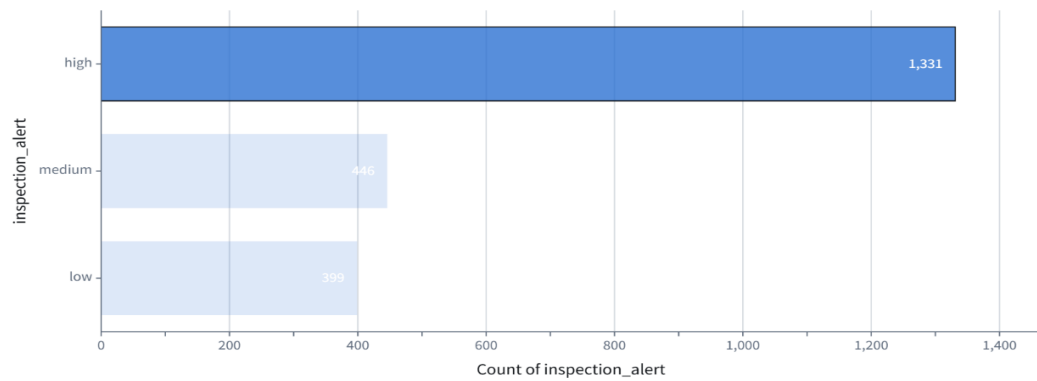
- In your dashboard, you may see the filters applied to each board as you selected or defined them in your analysis. These are interactive for viewers of the dashboard.
- Explore your dashboard parameters. Click the (x) to view the **Parameters**
- In the **Plant** filter, select **Plant 2**
- Click **Apply** at the bottom of the Parameters panel
  - The boards in the dashboard will dynamically update based upon your selection.
- Clear the parameter
- Click the **X** in the **Plant** text field to remove the list of plants -> click apply

## Inspection Prioritization Dashboard

Tab 1

+ New tab

### Inspection Alerts Histogram



Keep rows where inspection\_alert is high

Overridden

Show data

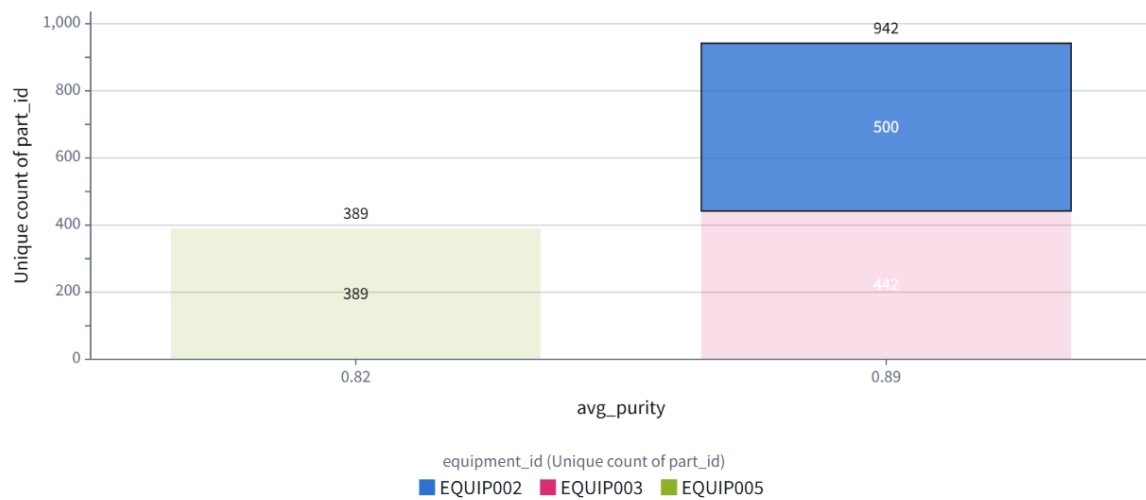
Affects 2 boards

### Count of Parts by Plant

Displaying 3 rows

| equipment_plant | equipment_id | count_distinct_of_part |
|-----------------|--------------|------------------------|
| Plant 2         | EQUIP002     | 500                    |
| Plant 3         | EQUIP003     | 442                    |
| Plant 5         | EQUIP005     | 389                    |

### Count of Parts by Equipment and Average Purity

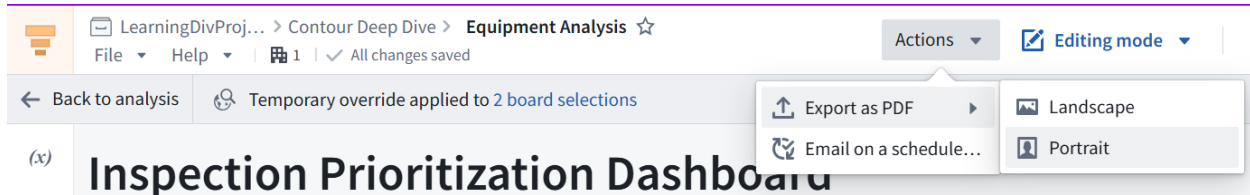




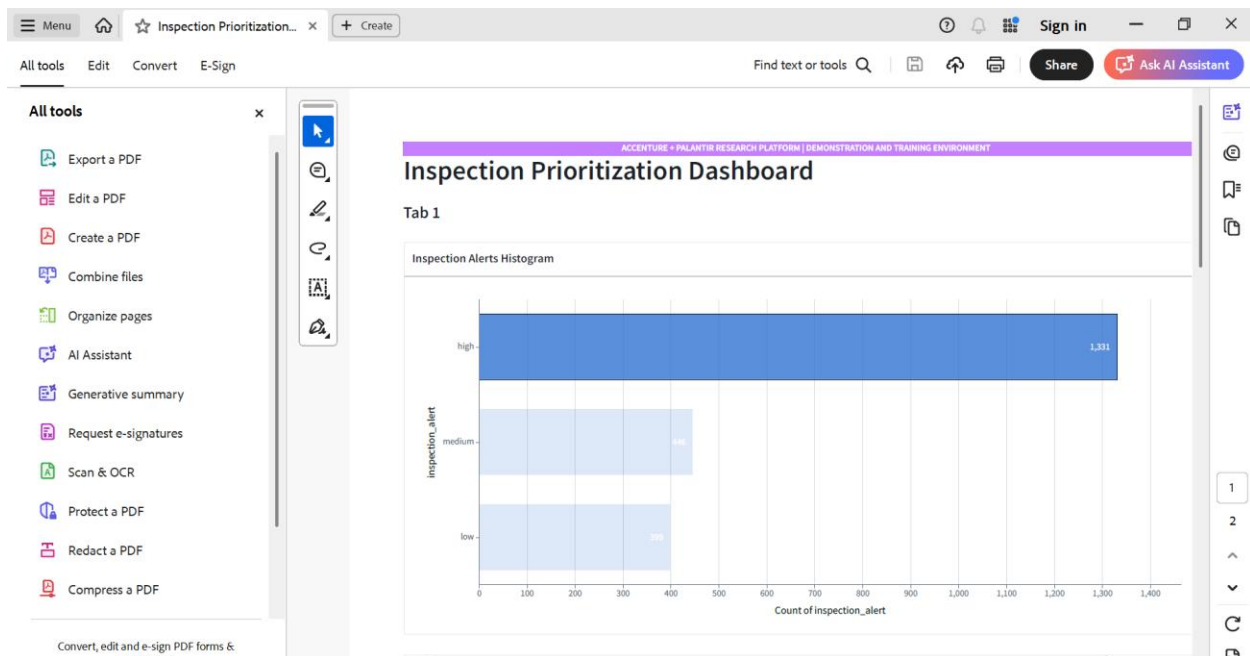
## C. Export the Dashboard

- In the top right of the screen, click Actions > Export as PDF > Portrait
- It will take Foundry a few seconds to generate the PDF.

Any filters and overrides applied in the dashboard are included in the PDF report.



Open the exported PDF.



----- Happy Learning! -----